Algorithm Design and Analysis

01 / Analysis and structures → 02 / Graph algorithms → 03 / Greedy proofs → 04 / Divide and randomize

01 / Name the guarantee

Worst-case maximizes over inputs; expected averages over stated randomness; amortized bounds the worst-case total of any sequence, divided by its length (no probability). Input size and arithmetic cost model must be explicit before interpreting any bound.

02 / Representation tradeoffs

Array indexing is O(1), interior shifts O(n). Doubly linked edits at known positions are O(1), but locating an index is O(n). Include both search and modification costs.

03 / BST versus AVL

BST search and update cost O(h); range reporting costs O(h+k). AVL maintains child-height difference at most one, ensuring h=O(log n). k is output size.

04 / Heap priorities

A min-heap keeps each parent no larger than its children. Minimum is O(1); insert and remove-min are O(log n). A heap array is not fully sorted.

05 / Hash-table assumptions

With suitable uniform hashing, chaining lookup is expected O(1+α), α=n/m. Linear probing needs tombstones or cluster repair after deletion. Cuckoo insertion can trigger expensive rebuilding.

06 / Graph traversals

Adjacency-list BFS and DFS cost O(n+m). BFS minimizes edge count. In undirected DFS, the tree edge from parent u to child v is a bridge iff low[v]>disc[u]; skip only that parent edge (by edge id).

07 / Path versus tree

Dijkstra relaxes d[u]+w(u,v) and requires nonnegative weights. Prim uses w(u,v) to connect a new vertex. MSTs minimize total network weight, not source-to-vertex distances.

08 / Greedy proof certificates

Fractional knapsack uses density and equal-weight exchange. Interval partitioning matches overlap depth. Huffman combines two minimum frequencies. Each proof depends on its specific feasible-choice structure.

09 / Recurrences

T(n)=T(n/2)+Θ(1) gives Θ(log n); 2T(n/2)+Θ(n) gives Θ(n log n); 3T(n/2)+Θ(n) gives Θ(n^log₂3). Include split and combination costs.

10 / Sort versus select

Randomized quicksort sorts both sides in expected O(n log n). Quickselect follows one side in expected O(n). Both can take $\Theta(n^2)$ on unlucky pivot sequences.

11 / Random structures

Fisher-Yates chooses j uniformly in {0,…,i} before fixing position i. Skip-list promotion probability one half gives expected O(log n) operations and expected O(n) storage.

12 / Read the input length

For q encoded in k≈log₂q bits, √q≈2^(k/2). A polynomial in the numerical value may be exponential in bit length; this distinction matters for cryptographic complexity.